Changes proposed to the PET CAEN DAQ for the Citiroc bench calibration

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1 Context

The CRYSP bench studies monolithic BGO and CsI(Tl) crystals, 48×48 mm^2 in cross section, read out by an 8×8 array of Hamamatsu S14160-6050HS SiPMs (6×6 mm^2, about 2 nF each). One of the readouts under study is the Citiroc 1A on a CAEN A5202, read through a DT5215 concentrator with the PET CAEN DAQ written by José María Benlloch (pet-caen-daq). Before taking data with a crystal and a source we calibrate the electronics with the SiPMs alone, in three short studies: the pedestal and noise of every channel with the SiPMs biased below breakdown; the delay between the trigger and the maximum of the shaped pulse, measured by scanning the hold delay; and the single-photoelectron (SPE) spectra of the dark pulses, which give the gain of every channel against the overvoltage and the shaping time. The results feed a simulation of the full chain, from the light in the crystal to the charge recorded by each readout, and the simulation is compared with the measurements channel by channel.

2 Why track and hold

The Citiroc samples its slow shaper in one of two ways. In the peak detector mode (FERSlib calls it PEAK_STRETCH) the chip follows the shaper from the trigger to the hold and keeps the maximum. In the track and hold mode it keeps the value at the instant of the hold. The peak detector is the right choice for the energy runs, where the pulse is a scintillation flash and the maximum is what matters. For the calibration it has two drawbacks with our sensors. The dark rate is several MHz per channel and the single-avalanche pulse of a 2 nF device falls in about 100 ns, so a window of a few hundred nanoseconds often contains a second dark pulse, and the maximum of the window picks it up. And the maximum of the noise over a window is not the noise of one sample, so the pedestal width measured in peak detector mode cannot be compared directly with the noise of the model. In track and hold the value is taken at a fixed time after the trigger: scanning the hold delay maps the shaped pulse point by point, the SPE spectrum is taken at the maximum of that pulse without the rest of the window, and the pedestal width is the noise of one sample. We intend to take the calibration in both modes, track and hold to understand the chain and peak detector because it is the mode of the energy runs.

3 What the DAQ does today

Reading the code of pet-caen-daq (main at 75086b2, 27 July 2026) next to FERSlib 5.0.1 we found four things relevant to these measurements.

1. **The mode cannot be selected.** FERSlib reads `PeakDetectorMode` from the configuration file and writes it into the `CitirocConfig` register; the JANUS GUI does not list it.

The DAQ validates every assignment against a catalogue and rejects the whole configuration if it contains an unknown setting, so a run with `PeakDetectorMode` does not start, and every run is taken with the peak detector.

2. **The Citiroc mode register is never written.** With automatic ASIC configuration the FPGA builds the slow-control stream of each Citiroc from board registers, and the chip modes come from `CitirocConfig` (`0x01000108`). FERSlib composes this register on every configuration. The DAQ writes it only during a threshold staircase, and restores the previous configuration afterwards, so a data run keeps whatever value the board held before. As a consequence `FastShaperInput` and `HV_Adjust_Range` are validated and audited as applied but do not reach the chip.

3. **The hold-delay scan keeps coarse spectra.** Each point of the scan stores the high-gain spectrum of every channel in 512 bins of 16 ADC codes (32 in the 14-bit range), the binning of the JANUS heatmap. This is enough to locate the maximum of the pulse, but the SPE peaks at the default gain are separated by only a few such bins, too few to fit their positions and widths, and the low gain is not kept.

4. **The recorded Citiroc stream is the planned one.** The HDF5 file of a run stores the 1144 bits of each Citiroc in `citiroc_stream_words`, but these are computed from the configuration, not read from the chip. The code to read the stream back and compare it (`VerifyCitirocReadback`) exists and is tested, but is not called in production. The file therefore records what the DAQ intended, not what the chip holds.

4 The pull requests

Each change is a separate pull request against jmbenlloch/pet-caen-daq, in English, with its documentation, unit tests, simulator integration tests, and the complete suite that runs without hardware (Go plain, HDF5 and integration tests, frontend format, lint, type check and unit tests). None of them changes the behaviour of a configuration that does not use the new settings, beyond making the existing settings take effect.

PR	change	compatibility and tests
#73	write `CitirocConfig` on every configuration, composed as FERSlib does	with JANUS defaults the register is `0x00070f21`; the production sample gives `0x00070fa1`; `FastShaperInput` and `HV_Adjust_Range` now take effect; register, stream and simulator readback tested
#74	add `PeakDetectorMode` (PEAK_STRETCH or TRACK&HOLD), built on #73	omitted means PEAK_STRETCH and gives the register of #73; register bits 2 and 3, stream bits 306 and 307, audit and frontend catalogue tested
#75	store full-resolution spectra of both gains in hold-delay scans, split by the channel's discriminator flag	additive table `hold_delay/full_spectra`; the 512-bin heatmap, the API and the UI are unchanged; rebinning the new spectra reproduces the heatmap exactly

A fourth change was planned and is not yet submitted: reading the Citiroc stream back after every configuration, storing it in the HDF5 file next to the planned one, and reporting any difference in the configuration audit. Testing it properly needs the simulator to build the stream from the board registers as the firmware does, and the read stream has to travel from the configuration step to the run file. We would rather agree on the design first.

5 What we need to use it

The changes help only once they run on the bench. After review and merge the container has to be rebuilt and installed on the laboratory computer. The first run in TRACK&HOLD is also the first hardware test of the register bits, which are confirmed from the FERSlib source but have not been exercised by this DAQ: a hold-delay scan with the dark pulses as signal should then show the held amplitude follow the shaped pulse, rising and falling with the delay, instead of the plateau that the peak detector gives. Until the merge the bench measurements are taken in peak detector mode, which the calibration procedure supports.

6 Questions for José María

1. The DAQ does not expose `EnableQdiscrLatch` and `EnableChannelTrgout`; #73 uses the FERSlib defaults (both 1), which changes stream bits 258 and 1143 with respect to the planned stream stored until now. Is that the value wanted in production?
2. Was the absence of `CitirocConfig` from the configuration plan deliberate, for a hardware reason we may have missed?
3. Is there a reason not to call `VerifyCitirocReadback` after configuration, and how would you like the read stream to be stored?
4. The schema version of the hold-delay artifact is left at 1, since #75 only adds a table. Would you prefer to increase it?